

## Biology

Date: 14-1-24

Ans 1. (4) It passes neither through our liver nor pancreas

Ans 2. (2) Living beings consume oxygen during respiration

Ans 3. (2) Bud develops due to repetitive cell division at a specific site

Ans 4. (4) Laced only

Ans 5. (4) Test tube B as pepsin can breakdown proteins into simple molecules only in acidic conditions

Ans 6. (1) Both A and R are true and R is the correct explanation of A.

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Ans 8. Rabbit and Deer belong to the same trophic level.

Ans 9. Oxygenated blood - Left side

Deoxygenated blood - Right side

Ans 10. (a) Yellow seed - Dominant character  
Green seed - Recessive character

(b) Round seed - Dominant character  
Wrinkled seed - Recessive character.

Ans 11. (a) If the male gamete has 24 chromosomes, female gamete will also have 24 chromosomes.

But, the total no. (48) will be halved, as a result of which ~~zygote~~ zygote will also have 24 chromosomes.

(b) (i) XX

(ii) XY

Ans 12. (i) Pancreas secretes both digestive enzymes and hormones.

(ii) If intake of iodine in our diet is low, the ~~thyroid~~ thyroid gland will not function properly as a result of which, thyroxine will not be released. This can lead to goitre.

Ans 13. (i) Pea plant have large number of contrasting features and therefore, Mendel selected pea plant.

(ii) Dominant traits - The traits which occur in maximum section of population and can express itself in both homozygous and heterozygous conditions.

Recessive traits - The traits which do not occur in maximum part of the population ~~can~~ and can only express themselves in homozygous conditions only.



(iii) The two visible features were:

- i) Height of plant
- ii) Size and colour of seed.

Ans 14. (a) (i) Ovary releases the female egg (ova) which is necessary for fertilization.

(ii) Oviduct or fallopian tube is the site of fertilization

(b) (i) When it has to receive a zygote, it thickens its walls and blood vessels.

(ii) When no fertilization takes place, the thick lining of uterus breaks down along with blood from the blood vessels.

(c) Placenta is a disc like structure embedded at the wall of uterus which provides the nutrition to the embryo fetus during pregnancy.